



Even experienced phone users drive worse while texting – A driving simulator study

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ARTICLE INFO

Article history:

Received 16 July 2020

Received in revised form 3 February 2021

Accepted 8 February 2021

Available online 8 March 2021

Keywords:

In-vehicle distraction

Texting

Mobile phone

Lane change task

Experience in mobile phone usage

ABSTRACT

Several studies have clearly shown that texting on a mobile phone increases crash risk (e.g. Dingus et al., 2006; Victor et al., 2014). However, the frequency of texting while driving still remains high (e.g. Vollrath, Huemer, Teller & Likhacheva, 2016). One reason may be that drivers are convinced that they are able to manage this dual task due to their competency in driving and texting. From a theoretical point of view, this may be true within limits – two well-learned, automatically processed tasks may require so few resources most of the time that interferences may not occur.

In order to test this assumption, a study with a standardized driving simulator task (the lane change task, LCT; Mattes, 2003) was done with $n = 40$ drivers selected for their driving experience and tested for their texting abilities. The effect of driving experience (high vs. low) and texting competence (high vs. low) on driving performance was examined in single (driving only) and dual-task (driving and texting) conditions. Additionally, a subjective assessment of one's task performance was obtained. Texting on the mobile phone significantly impaired driving performance. While driving experience did not have an influence, the deteriorating effect of texting was significantly less pronounced in highly competent phone users. Interestingly, this objective effect was not found in the subjective assessment. Drivers with a high texting competence felt as impaired as drivers with a low texting competence. This is in line with the finding that even in this simple driving task their performance was still significantly deteriorated as compared to driving, only. Thus, it seems that the reason why people text while driving is not that they are not aware of the performance loss. However, this awareness of the possible risk does not seem sufficient to prevent them from texting while driving.

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1. Introduction

Quite a number of naturalistic driving or accident risk studies have found a strong increase of crash risk due to distraction. Texting on a mobile phone has the strongest and most consistent negative effect (e.g. Dingus et al., 2006; Victor et al., 2014; Young, 2017). Meta-analyses on experimental studies of secondary tasks while driving support this finding by showing that the negative effects of secondary tasks depend on the type of the task (e.g. Caird et al., 2014; Ferdinand & Menachemi, 2014; Oviedo-Trespalacios, Haque, King & Washington, 2016; Papantoniou, Papadimitriou & Yannis, 2017; Vollrath, Huemer,

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Nowak & Pion, 2014). In accordance with the multiple resources model of Wickens (2002) tasks using the visual channel to process spatial information and including manual reactions interfere most with the driving tasks as this also requires these resources. The secondary task and the driving task compete for these resources. As these are limited, performance decrements result. Looking at the interference due to texting while driving a bit more closely, information from the environment relevant for driving is simply not available to drivers looking at their smartphone. Thus, they fail to notice that drivers in front are braking (rear-end crashes, see Victor et al., 2014) or that they are leaving the lane (leading to single vehicle crashes).

Thus, at least for texting on the mobile phone it should be obvious for every driver that this is detrimental for driving. However, observational studies have found large percentages of drivers texting when driving (e.g. Asgharabad, Tahami & Khanjani, 2013; Pickrell and Li, 2016; Vollrath et al., 2016). One reason for this may be that while drivers are aware that visual distraction is dangerous in general, they underestimate the actual risk for themselves (Huemer & Vollrath, 2011). In addition, to make texting while driving seem safer, drivers may also adapt their behavior by only engaging in secondary tasks in relatively low risk driving situations (e.g. Huth, Sanchez & Brusque, 2015; Metz, Landau & Hargutt, 2015) or by driving more slowly and with a longer headway (e.g., Fitch, Grove, Hanowski & Perez, 2014; Oviedo-Trespalacios, Haque, King & Washington, 2016).

This idea of compensation is also in line with the multiple resources theory. While resources are limited they may also be shared. If both tasks require only a small part of the available resources, as in texting while driving on an easy stretch of the road, it may be possible that both tasks can be done concurrently without interference. Thus, combining easy driving situations with easy secondary tasks may work well even when both use the same resources. This point has also been made by Shiffrin and Schneider (1977) in their theory of automatic and controlled human information processing. If tasks are very well learned they can be performed automatically and do not require many processing capacities. This idea was also extended in a review by Engström et al. (2017) for secondary tasks which primarily increase cognitive, but not visual load. In their review they found that cognitive load only impairs subtasks of driving that require cognitive control, but not subtasks with automatic performance. Finally, in their integrative review of dual-task and task-switching research, Koch, Poljac, Müller and Kiesel (2018) found three processes which could lead to practice effects in dual-task performance one of which was task automatization.

If this idea is transferred to the effect of secondary tasks while driving, a new moderating factor is introduced for both the driving task and secondary tasks. With regard to driving, one would expect that driving becomes more and more automatic the longer and more one drives. Drivers with a longer driving experience and drivers with a larger mileage should mainly use automatic information processing while driving. For texting on the mobile phone, similar effects would be expected. People who text all the time and have done that for several years should also mainly use automatic information processing while texting. Thus, experienced drivers and competent mobile phone users should be best able to text while driving without losses in either the primary or secondary task. A study by Rumschlag et al. (2015) compared skilled and not skilled texters in a driving simulation and indeed found differences: For the not skilled texters, lane excursions were the larger the longer the text messages were. Skilled texters did not show this effect (perhaps because their text messages were short and with less variance), but a strong age effect occurred. The older the drivers were, the more lane excursions occurred. One possible reason for this effect could be the decrease of resources with age which then leads to an increase of impairment. However, driving experience was not examined in this study. Thus, the combined influence of driving experience on the one hand and texting experience on the other has still not been examined to our knowledge.

In addition, it would also be interesting to examine the effect of specific experience of texting while driving, as that may be a skill in itself. However, as texting while driving is forbidden in Germany we assumed that it would be difficult to obtain true and realistic answers to the question of how often one texts while driving. In order to avoid this bias we decided to focus on experience with texting and with driving, separately.

This idea was tested in a driving simulator study using the lane change task (Mattes, 2003; ISO/DIS 26022, 2010) as a simple driving simulator and comparing high and low experienced drivers and high and low competent mobile phone users while driving only and when texting while driving. Additionally, it was examined to what extent drivers can assess their performance decrements due to texting while driving. As described above, our hypotheses were that experienced drivers and experienced texters would both show better dual-task performance (driving and texting) than inexperienced drivers or texters. Both effects should also add up so that experienced drivers who were also experienced texters should have the least negative effects of texting while driving. The method is described in the next section.

2. Method

2.1. Experimental design

The basic aim of the study was to examine the effect of practice in driving and texting on the phone on the ability to combine both tasks and to, especially, drive without losses in driving performance.

To this aim, the study used a 2×2 quasi-experimental design comparing the two independent variables 'driving experience' (low vs. high) and 'texting competence' (low vs. high). A quasi-experiment was used as the two independent variables could not be actively varied. From this design, four independent groups resulted which are shown in Table 1.

Table 1

Experimental design including the number of subjects in the different conditions.

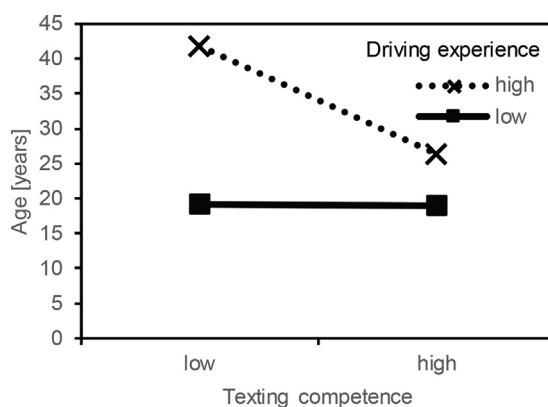
		Texting competence		
		low	high	sum
Driving experience	low	7	13	20
	high	13	7	20
	sum	20	20	40

For the driving experience, this was included in the selection of the subjects. For drivers with a low driving experience we looked for subjects driving a maximum of 5000 km per year and having their driver's licence for a maximum of three years. For the highly experienced group, drivers were supposed to drive at least 10,000 km per year and have their driver's licence for at least five years. 20 drivers were selected for each group.

To select for texting competency, a simple test was done using the personal mobile phone of the subject. Every driver was told to type a given text (79 words). They were supposed to do this as fast as possible and with so few errors as possible. The time required and the number of errors were measured. When looking at the distribution of time and errors it was obvious that subjects used different strategies. Some typed very rapidly, but had many errors. Others took their time, but had hardly any errors. Thus, for the selection of texting competency, a combined measure of errors and time was used. Subjects took from 98 to 492 s to complete the text and did 0–16 errors. For the highly competent texting group, subjects were allowed up to 192 s to type the text while doing less than seven typing mistakes. Thus, in the highly competent group the mean time was 147 s and 2.1 errors. The less competent group took 261 s and had 3.5 errors. As shown in Table 1, this resulted in 20 subjects in each group. However, there were more highly competent texters in the low driving experience group and less in the high driving experience groups.

In an experiment, the four groups would have been randomly assigned thus controlling for other confounding factors. As this was not possible in this quasi-experiment, we looked at possible confounders, the most important being age. Fig. 1 shows the mean age in the four groups. For drivers with a low driving experience, the two groups of differing texting competency were of the same age. However, in the group with high driving experience, the age of the drivers with a low texting competency was 42 years as compared to 26 years for the drivers with a high texting competency. Thus, if a main effect of texting competency was found in the group with high driving experience, this could also be due to the older age of the less texting competent drivers.

In order to measure the effect of texting on driving, the lane change task (LCT; Mattes, 2003) was used. This is a simple driving simulation which runs on a desktop computer using a gaming steering wheel and pedals to steer a simulated car. The main driving task is to drive on a three-lane road. About every 150 m a sign becomes visible which indicates the current target lane. The driver is instructed to change the lane as soon and fast as possible when the sign becomes visible. The driver is instructed to drive with a constant speed of 60 km/h. Driving faster is prevented by the simulation. Thus, drivers are instructed to just go as fast as possible. One trial consists of 18 lane changes with a randomized order of these lane changes. The recommended measure of driving performance is the mean of the deviation from an ideal (artificial) trajectory. This assumes the driver to stay at the middle of the lane and then make a lane change at 30 m before the lane change sign, taking 10 m to change the lane. This measure was also used in the present study. Additionally, the driving task in the lane change task basically comprises lane changes (in reaction to respective signs) in alternation with lane keeping. Drivers might try to synchronize their texting with the driving task by focusing on a lane change and then do the texting in the meantime until the next lane change was required. In order to examine that, the mean deviation was also computed for lane change and lane

**Fig. 1.** Average age in the four groups combining texting competence and driving experience.

keeping parts of the task, separately. The lane change phase was set to begin when the sign became visible (40 m away from the sign) until 60 m after the sign. The next 40 m were then regarded as lane keeping phase.

This task was first learned by all subjects by doing 4 practice trials. After that, every subject drove two trials with and two further trials without texting. The order of these four trials was randomized. The results from the two trials without and with texting were averaged to obtain more stable parameters. Thus, ‘texting’ was introduced as an additional independent variable (with texting vs. driving only).

With regard to the texting task, the experimenter wrote a text message every 20 s to the subjects containing simple questions like “What kind of weather is today?”, “Which was the last book you have read?”, “What is your favourite drink”, etc. A complete list of the 18 questions used can be obtained from the authors on request.

Besides the driving parameter ‘mean deviation from ideal lane’, every driver was asked to evaluate the own performance after each trial. School grades were used ranging from 1 (very good) to 6 (very bad) in order to examine how well drivers can assess their performance.

2.2. Participants

Table 1 shows the distribution of the 40 drivers in the four groups. 10 of them were male and 30 females. The mean age was 28 years ($sd = 12.6$). Overall, they had their driver's licence for a mean of 120 months ($sd = 143$) and drove on average between 5000 and 10 000 km per year.

2.3. Procedure

After arrival, subjects were instructed that the study was concerned with examining the effects of using a mobile phone to type messages while driving in a simple driving simulator. They were also told that we were interested in the effects of practice with typing and did a simple typing test. The results were used as assessment of typing competence (see above). After obtaining sociodemographic data subjects were instructed in the lane change task. They did four trials of this task in order to achieve a well-trained performance level. After that, the experimental trials began. In the two trials without distraction they were told to do the lane change task as well as they could. In the other two trials they were instructed to answer text messages where the examiner asked simple questions about daily events (e.g., “Did you drive with a bus today?”) which could be answered with a short text message. Drivers were instructed to answer as many of these questions as they could. However, driving the lane change task was still instructed to be the main task which should not suffer by the texting. After finishing, some questions concerning phone use while driving were given.

3. Results

A $2 \times 2 \times 2$ mixed ANOVA was computed for the mean deviation from the ideal lane with ‘texting’ as within-subjects factor and ‘texting competence’ and ‘driving experience’ as between subject factors. The same ANOVA was also used for the lane keeping phase and the part of the lane changes. Table 2 gives the results. For the overall mean deviation, there was a strong main effect of texting, a second main effect of texting competence and an interaction between texting and texting competence. Additionally, there was a tendency ($p = 0.054$) for an interaction between driving experience and texting competence. Similar effects were found for the lane keeping phase with the exception of the interaction, which was not significant. In contrast, the main effect of texting competence was not as strong while changing the lane, but the interaction of driving experience and texting competence was clearly significant while changing the lane.

As the weak interaction between driving experience and texting competence in the overall mean deviation from the ideal lane seems to be due to the lane change phase (significant interaction there), Fig. 2 gives the mean deviation while changing the lane. It shows that only drivers with a high driving competence and a high texting competence perform better than drivers with a high driving competence but with a low texting competence. Thus, texting competence has a positive effect on

Table 2

Results of the $2 \times 2 \times 2$ mixed ANOVA with the mean deviation from the ideal lane as dependent variable. The first part gives the overall mean deviation, followed by the mean deviation during lane keeping and while changing the lane.

Effect	Mean deviation from the ideal lane								
	Overall			During lane keeping			While changing the lane		
	F(1,36)	p	Eta	F(1,36)	p	Eta	F(1,36)	p	Eta
Texting (T)	79.4	0.000	0.688	30.1	0.000	0.456	68.0	0.000	0.654
Driving experience (DE)	1.3	0.264	0.034	1.4	0.233	0.039	0.8	0.371	0.022
Texting competence (TC)	5.3	0.028	0.128	6.7	0.014	0.158	3.5	0.067	0.090
Texting $\times$ Driving experience	0.2	0.653	0.006	0.4	0.527	0.110	0.0	0.989	0.000
Texting $\times$ Texting competence	13.7	0.001	0.276	7.7	0.009	0.177	9.4	0.004	0.207
Driving experience $\times$ Texting competence	3.9	0.054	0.100	0.6	0.431	0.017	4.5	0.040	0.112
T $\times$ DE $\times$ TC (three-way interaction)	0.5	0.475	0.014	0.5	0.475	0.014	0.2	0.649	0.006

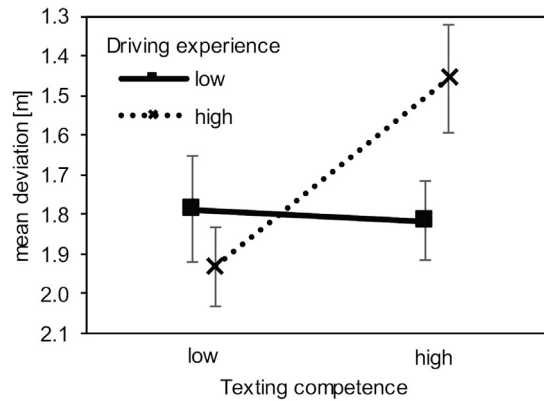


Fig. 2. Effects of texting competence and driving experience on the performance in the lane change task (mean and standard error of the mean deviation from ideal lane during the lane change phase). As high values correspond to a bad performance the y-axis has been inverted to show bad performance at the bottom of the graph.

the performance only for drivers with a high driving experience. For drivers with a low driving experience, no effect of texting competence could be shown. Thus, overall drivers with a high driving experience and high texting competence perform best in the lane change task. As shown in Fig. 1, these were also the younger drivers within the group of higher driving experience. So this effect may be partly due to the young age of the drivers.

However, the main effect of texting and the interaction of texting and texting experience is not influenced by this interaction, as no three-way interaction was found. As Fig. 3 shows, texting increases the mean deviation from the ideal lane. However, this effect is much stronger in the group with low texting competence. As post-hoc *t*-tests with a Bonferroni adjusted alpha of $0.05/4 = 0.0125$ (four *t*-tests) show, both groups do not differ in their performance when driving only ($t_{38} = 0.7$, $p = 0.481$), but differ significantly when texting and driving ($t_{38} = 2.8$, $p = 0.008$). When examining the effect of texting separately for both groups using a within-subjects *t*-test, texting impairs performance in the low competence group ($t_{19} = 8.4$, $p < 0.001$), but also in the high competence group ($t_{19} = 5.0$, $p < 0.001$). Thus, texting competence reduces the negative effects of texting on driving, but some (significant) impairment remains.

This effect is found during the lane change phase as well as during the lane keeping phase. Thus, there is no indication of a strategy of the drivers to adapt the secondary task performance to the requirements of the driving task by, e.g., texting only when keeping the lane but focusing on driving while changing the lane. If this had been the case, there should be a much smaller effect of texting (and perhaps a smaller interaction) when changing the lane, but a stronger effect during lane keeping. However, this could not be shown.

The same analysis was done for the self-assessment of driving performance. In this 2x2x2 mixed ANOVA, the only significant effects were that of texting ($F_{1,36} = 335.5$, $p = 0.000$) and of texting competence ($F_{1,36} = 4.4$, $p = 0.043$). All other main effects and interactions were not significant (all $p > 0.05$). As Fig. 4 shows, drivers are very well aware of the negative effects of texting on driving performance, regardless of being low or high competent in texting. However, it is interesting to see that drivers who are well able to text assess their driving performance both with and without texting as better than those with a low texting competence. In comparison to the objective results, drivers with a high texting competence do not appear to be

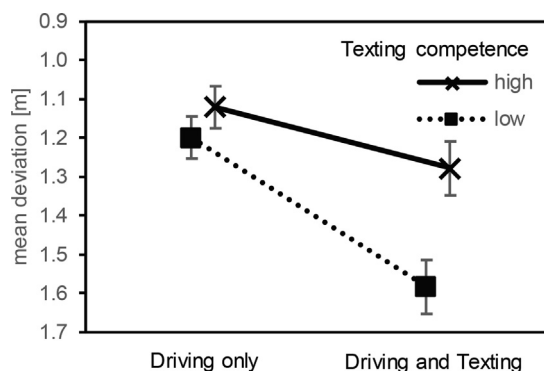


Fig. 3. Effect of texting for drivers with a low and high texting competence (mean and standard error of the mean deviation from ideal lane). As high values correspond to a bad performance, the y-axis has been inverted.

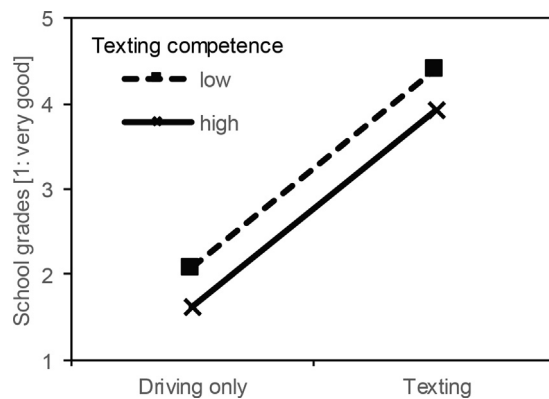


Fig. 4. Self-assessment of driving performance using school grades (1: very good, 6: very bad) after driving only and after driving with texting in the two groups with low and high texting competence.

aware of the fact that they are indeed able to cope somewhat better with texting and driving than others with a low texting competence.

4. Discussion

In accordance to the theory of [Shiffrin & Schneider \(1977\)](#) well-trained phone users who can text fast and with few errors are not impaired as much by texting while driving as compared to not-so-competent phone users. Their handling the phone is probably mainly done by automatic information processing, thus requiring less resources which does not interfere with driving as much as for less competent phone users. From a behavioural point of view, they may simply type faster and are thus distracted from driving for shorter periods of time. As we were not able to record the time spent typing this interpretation could not be tested in the present study. However, although the deviation from an ideal lane position during texting was less than for the less competent phone users, there was still a significant deterioration as compared to driving without distraction. As the driving simulation uses a very limited driving situation, it is hard to predict the effects of phone use competency for crash risk. However, the lane change task is a well-established tool for assessing safety-relevant driving behaviour. Moreover, when the effects are significant in such a relatively easy driving task they could be even stronger in real driving in complex and demanding situations. However, it could also be that in real driving compensatory behaviour (only texting on easy parts of the route) is more likely than in the driving simulator study. Thus, replicating this effect in real driving or at least in more complex driving situations in the driving simulator would be very interesting.

As there was some confounding between age and texting competence, it would also be fine to replicate this study with a more balanced design. Additionally, it would be very interesting to also use a group with a high texting competency while driving. This would require to find drivers who are willing to confirm that they often text while driving all though this is forbidden in Germany. However, comparing this to texting competency per se one could examine whether texting while a driving as a dual task is a competency per se. This would contribute to the discussion in dual task research whether dual task performance is a special competency which can be trained.

In order to avoid the difficulty of finding drivers willing to confirm that they are texting while driving one could also use naturalistic driving data to select similar drivers (e.g., with regard to sex, age, driving experience, annual mileage) who either text frequently while driving and who do it only infrequently. One would then have to select similar driving situations where both groups of drivers are texting and obtain some measure of driving performance, e.g. standard deviation of lane position as an indicator of lane keeping performance. While this would be a very realistic and valid approach, it could be difficult to control for all confounding factors like traffic situation, traffic density, environment etc.

With regard to driving experience, one would also expect a smaller effect of texting for experienced drivers. However, this was not the case. One reason could be that experience in real driving does not transfer to driving in the driving simulator. However, at least for drivers with a high texting competency (younger drivers) those with a larger driving experience performed much better in the driving simulator than those with a smaller driving experience. This result supports the validity of the simulation. However, this did not affect the impairment due to texting while driving which was similar regardless of driving experience. From this viewpoint it does not make sense to have stronger regulations for younger drivers or to be more lax for experienced drivers. Or, to put it differently, the problem is more strongly related to the secondary task, not so much to the primary task of driving. It could also be that the driving task was just too easy so that even less experienced drivers were easily able to perform well (floor effect). Thus, it would be interesting to see if an effect of driving experience was found in complex traffic situations with a more difficult driving task which should present few problems for the experienced drivers but require a large amount of processing resources for less experienced drivers.

When asking the drivers directly after their trip about the quality of their driving performance all drivers were well aware of the loss of driving performance due to texting on the mobile phone. This may partially be due to an answer bias of social desirability knowing that one should not text on the phone while driving. It was also interesting to see that highly competent phone users rated themselves somewhat better than not-so-competent users. However, the objectively smaller detrimental effect of texting in highly competent users was not found in the subjective assessment. This could be due to the answer bias mentioned. However, it could also mean that the self-assessment is not very precise and differentiated. Overall, these results show that the subjective belief that texting on the phone decreases driving performance is not sufficient to refrain drivers from doing so. This was also supported by an exploratory interview after the experiment where about 80% of the subjects indicated that they used the phone while driving at least from time to time.

Why do people do this although they know that they drive less safely? One reason could be that they think that the impairment is still within a range of safe behaviour. When texting while driving, one may react more slowly and one may swerve, but drivers do not experience a crash every time they use the phone. While this is simply due to our traffic being so safe and the absolute crash risk being so small, subjectively every phone use without a crash may be experienced as a reinforcement of the impression that one is still on the safe side. Additionally, the benefits and direct positive reinforcement from using the phone may be very large as compared to the risky, but highly improbable negative results of distraction. Thus, for prevention, simply demonstrating the negative effects of mobile phone use on driving performance does not seem to be sufficient to substantially influence mobile phone users' dangerous habits.

However, the study shows that even experience texters show a significant negative effect of texting on driving performance. Presenting these (and similar) results to drivers in the context of prevention might help to clarify that experience and caution is not always sufficient to counteract these negative effects of texting. Additionally, one could easily demonstrate these negative effects to the experienced texters and thus perhaps lower their impression of being able to counteract these negative effects.

However, as it seems to be a large need to interact by phone, perhaps the only solution is to further encourage the use of text-to-speech letting the phone read out the text messages and to use voice messages for the reply. This enables the driver to look at the road while being engaged in this kind of interaction. To this aim, apps could introduce a driving mode which could also be automatically be activated when the app detects that the user is driving.

With regard to the limitations of the study, besides using a simple driving simulator (see discussion above) the sample was limited to a mostly young, student, female population. This was mainly due to practical availability of subjects. However, mobile phone use and a high competence in using it is more frequent in young people. Moreover, negative effects may be more pronounced in inexperienced, young drivers with their higher crash risk. However, especially with regard to driving experience, it would be interesting to replicate this study with an older sample of drivers. However, it may be difficult to find older drivers with a comparable high texting competence as younger drivers.

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